

- 2 A common weed in barley fields is couch grass. A laboratory investigation was carried out to study the effect of couch grass on the yield of barley.

Barley was grown in standard sized pots in a glasshouse. The number of barley plants per pot was constant. The supply of water and mineral salts, the light and the temperature were controlled automatically. Six replicates of each of four different experiments were carried out.

experiment 1: Couch grass and barley were sown together at the same time.

experiment 2: Couch grass was sown 14 days before the barley.

experiment 3: Barley sown 14 days before the couch grass.

experiment 4: Barley grown alone.

The yield of barley from each of the different experiments was measured.

The results showed that the yield of barley from experiments 1, 3 and 4 were the same. The results from experiment 2 showed a significant decrease in the yield of barley in comparison to the others.

- (a) One hypothesis to explain the results of these experiments is:

Older couch grass roots produce a substance that inhibits the growth of younger barley roots.

- (i) Outline how this hypothesis could be tested.

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- (ii) Suggest **one other** hypothesis that could explain the results of experiment 2.

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- (iii) Suggest a null hypothesis that could be used in a statistical test to compare the yield of barley with the control.

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[1]

- (b) In a second investigation the biodiversity in two fields of barley was studied.

Barley was planted at the same density in each field and left to grow until it reached maturity.

In one of the fields pesticides (herbicides and insecticides) were used to manage pests.
 In the other field pesticides were not used.

When the barley was mature, a standard sampling method was used to estimate the number of each different type of organism found in each field. These numbers were used to calculate the species diversity of each field using the formula for Simpson's Index of Diversity (D).

$$D = 1 - \left(\sum \left(\frac{n}{N} \right)^2 \right)$$

n = number of individuals of each type present in the sample
 N = the total number of all individuals of all types
 Σ = the sum of

Table 2.1 shows the results of this investigation and some of the calculations for Simpson's Index of Diversity.

Table 2.1

type of organism	mean number of individuals in 10 m ² samples from fields using pesticides	$\left(\frac{n}{N} \right)^2$	mean number of individuals in 10 m ² samples from fields without pesticides	$\left(\frac{n}{N} \right)^2$
dicotyledons	3	0.004	60	0.021
monocotyledons	15	0.102	120	0.084
ground beetles	20		45	
butterflies and moths	4	0.007	35	0.007
bees	0	0.000	95	0.053
small mammals	3	0.004	23	0.003
birds	2	0.002	36	0.008
total	47		414	

- (i) Complete Table 2.1 **and** use the data to calculate D for both fields of barley.

In fields using pesticides $D =$

In fields without pesticides $D =$ [3]

- (ii) State what the results of this investigation show about the effect of pesticides on the plants and animals studied.

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